

The matching problem

Recall that if $E_1, E_2, \dots, E_n$ are events of an experiment, then the probability of occurrence of at least one of the events is,

$$P(E_1 \cup E_2 \cup E_3 \cup \dots \cup E_n)$$

$$= \sum_{i=1}^n P(E_i) - \sum_{i_1 < i_2} P(E_{i_1} \cap E_{i_2})$$

$$+ \dots + (-1)^{r+1} \sum_{i_1 < i_2 < \dots < i_r} P(E_{i_1} \cap E_{i_2} \cap \dots \cap E_{i_r})$$

$$+ \dots + (-1)^{n+1} P(E_1 \cap E_2 \cap \dots \cap E_n)$$

Problem: Suppose that each of N men at a party throws his hat into the center of the room. The hats are first mixed up and then each man selects randomly selects a hat. What is the probability that none of the men selects his own hat?

Solution:

Let E_i be the event that the i^{th} man selects his own hat

The desired probability is

$$P(E_1^c \cap E_2^c \cap E_3^c \cap \dots \cap E_N^c)$$

$$= P\left(\left(\bigcup_{i=1}^N E_i\right)^c\right)$$

$$= 1 - P\left(\bigcup_{i=1}^N E_i\right).$$

Now we compute $P\left(\bigcup_{i=1}^N E_i\right)$

using the formula

$$P\left(\bigcup_{i=1}^N E_i\right)$$

$$= \sum_{i=1}^N P(E_i) - \sum_{i_1 < i_2} P(E_{i_1} \cap E_{i_2})$$

$$+ \dots + (-1)^{r+1} \sum_{i_1 < i_2 < \dots < i_r} P(E_{i_1} \cap E_{i_2} \cap \dots \cap E_{i_r})$$

$$+ \dots + (-1)^{N+1} P(E_1 \cap E_2 \cap \dots \cap E_N)$$

If we regard the outcome of this experiment as a vector of N numbers, where the i^{th} element is the number of the hat drawn by the i^{th} man, then there are $N!$ possible outcomes

For instance $(1, 2, 3, \dots, N)$ means that each man selects his own hat.

Now, the event

$E_{i_1} \cap E_{i_2} \cap \dots \cap E_{i_r}$ denotes

that each of the r men
selects his own hat, the

remaining $(N-r)$ men can

select hat from $(N-r)$ hats

in $(N-r)!$ possible ways.

The probability

$$\begin{aligned} P(E_{i_1} \cap E_{i_2} \cap \dots \cap E_{i_r}) \\ = \frac{(N-r)!}{N!} \end{aligned}$$

Also, there are $\binom{N}{r}$ terms in

$$\sum_{i_1 < i_2 < \dots < i_r} p(E_{i_1} \cap E_{i_2} \cap \dots \cap E_{i_r}),$$

it follows that

$$\sum_{i_1 < i_2 < \dots < i_r} p(E_{i_1} \cap E_{i_2} \cap \dots \cap E_{i_r})$$

$$= \binom{N}{r} \frac{(N-r)!}{N!}$$

$$= \frac{N!}{(N-r)! r!} \frac{(N-r)!}{N!}$$

$$= \frac{1}{r!}$$

Thus,

$$P\left(\bigcup_{i=1}^N E_i^c\right)$$

$$= 1 - \frac{1}{2!} + \frac{1}{3!} - \dots + (-1)^{N+1} \frac{1}{N!}$$

The desired probability is,

$$P(E_1^c \cap E_2^c \cap \dots \cap E_N^c)$$

$$= 1 - \left[1 - \frac{1}{2!} + \dots + (-1)^{N+1} \frac{1}{N!} \right]$$

$$= \sum_{i=0}^N \frac{(-1)^i}{i!}$$

Problem: Compute the probability that if 10 married couples are seated at random at a round table, then no wife sits next to her husband.

Solution: Let E_i ($i = 1, 2, \dots, 10$) be the event that the i^{th} couple sit next to each other.

The desired probability

$$P(E_1^c \cap E_2^c \cap \dots \cap E_{10}^c)$$

$$= 1 - P\left(\bigcup_{i=1}^{10} E_i^c\right).$$

Now we compute $P\left(\bigcup_{i=1}^{10} E_i\right)$

by the formula

$$P\left(\bigcup_{i=1}^{10} E_i\right)$$

$$= \sum_{i=1}^{10} P(E_i) - \dots +$$

$$(-1)^{r+1} \sum_{i_1 < i_2 < \dots < i_r} P(E_{i_1} \cap E_{i_2} \cap \dots \cap E_{i_r})$$

$$+ \dots - P(E_1 \cap \dots \cap E_{10})$$

To compute $P(E_{i_1} \cap E_{i_2} \cap \dots \cap E_{i_r})$

we first note that there are $19!$ ways of arranging 20 people around a round table.

The number of arrangements that result in a specified set of ' r ' men sitting next to their wives is obtained by thinking of each of ' r ' married couples as being single entities.

In this case, we would need to

arrange $(20-r)$ people around a round table.

$$\begin{aligned}\text{There are } (20-r-1)! \\ = (19-r)!\end{aligned}$$

arrangements.

Since each married couple can be arranged next to each other in one of the two possible ways,

it follows that $2^r (19-r)!$

arrangements of r men sitting next to their wives.

So,

$$P(E_{i_1} \cap E_{i_2} \cap \dots \cap E_{i_r})$$

$$= \frac{2^r (19-r)!}{19!}$$

$$P\left(\bigcup_{i=1}^{10} E_i\right)$$

$$= \binom{10}{1} \frac{2^1 18!}{19!} - \binom{10}{2} \frac{2^2 17!}{19!}$$

$$+ \binom{10}{3} \frac{2^3 16!}{19!} - \dots - \binom{10}{10} \frac{2^{10} 9!}{19!}$$

$$= \underline{\underline{0.6605}}$$

Conditional Probability and Independence.

This is one of the most important properties in probability theory. The importance of this concept is two fold.

- ① Computing probability when some partial information is known.

② In some situations, even when partial information is not available, Conditional probabilities can be used to compute the desired probability

• Let E and F be events of an experiment. The "conditional probability that E occurs given that F has occurred" is denoted by $P(E/F)$.

- If the event F occurs, then in order for E to occur, it's necessary that the actual occurrence be a point in both E and F . i.e., $E \cap F$
Since F has occurred, it follows that F becomes our new sample space. and hence the probability that the event $E \cap F$ occurs is,
$$\frac{P(E \cap F)}{P(F)}$$

If $P(F) > 0$, then

$$P(E/F) = \frac{P(E \cap F)}{P(F)}$$

Problem: Suppose that two dice are tossed. What is the probability that sum of the two dice equals 8 given that the first die is 3?

Solution:

Let E be the event that the sum of the two dice equals 8.

F - event that the first die is 3 when two dice tossed

$$F = \{ (3,1), (3,2), (3,3), (3,4), (3,5), (3,6) \}$$

$$P(F) = \frac{6}{36} > 0.$$

$$E \cap F = \{ (3,5) \}$$

$$\Rightarrow P(E \cap F) = \frac{1}{36}$$

The desired probability

$$P(E/F) = \frac{P(E \cap F)}{P(F)}$$

$$= \frac{\frac{1}{36}}{\frac{6}{36}}$$

$$= \frac{1}{6}$$

Problem: A coin is flipped twice.

What is the conditional probability that both flips land on heads

given that

- (a) the first flip lands on heads.
- (b) at least one flip lands on heads.

Solution:

Let B - be the event that both flips lands on heads

$$B = \{ (H, H) \}$$

(a) Let F be event that the first flip is heads

$$F = \{ (H, H), (H, T) \}$$

$$P(F) = \frac{2}{4} = \frac{1}{2}$$

$$P(B \cap F) = P(\{ (H, H) \}) = \frac{1}{4}$$

$$P(B/F) = \frac{P(B \cap F)}{P(F)} = \frac{1}{2}$$

(b) Let A be the event that at least one flip lands on heads

$$A = \{ (H, H), (H, T), (T, H) \}$$

$$P(A) = \frac{3}{4}$$

$$P(B/A) = \frac{P(B \cap A)}{P(A)}$$

$$= \frac{\frac{1}{4}}{\frac{3}{4}} = \frac{1}{3}.$$

Problem: In a card game bridge the 52 cards are dealt out equally to 4 players - called E, W, N and S. If N, S have a total of 8 spades among them, then what is the probability that E has

3 of the remaining 5 spades?

Solution: We solve this by writing down a reduced sample space.

Given that both N and S have 8 spades among their 26 cards.

There remains a total of 26 cards, exactly 5 of them being spades to be distributed among E, W hands.

Since each distribution is equally likely, it follows that the conditional probability that E will have exactly 3 spades among E 's 13 cards is,

$$\frac{\binom{5}{3} \binom{21}{10}}{\binom{26}{13}} = 0.339.$$

